Gopher Invitational - Repaired practice edition - Answer key

Division B | Heredity | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: Which base pairs with guanine in DNA?

Unit: DNA structure, replication, and mutation.
Worked-case data: A DNA template strand reads 3′-TAC GGA TTC-5′. A mutation changes the third codon’s template to ATC. Classify the mutation.
Separate investigation observations:
Normal mRNA: 5′-AUG GCU UGG AAA-3′ codes Met-Ala-Trp-Lys.
Patient mRNA: 5′-AUG GCU UGA AAA-3′.
Protein size: The patient’s protein is much shorter than normal.

A. Adenine

B. Thymine

C. Uracil

D. Cytosine

Correct: D